

题1:

① $ab@c+*$

② $abcde/+*+$

③ $a@bc@*+d+$

④ $xy+z*0=ab+c\uparrow abc\uparrow\uparrow if$

题2:

2、写出下面赋值句 $A := B * (-C + D)$ 的自下而上语法制导翻译过程。给出所产生的三地址代码。

Syntax:

(1) $S \rightarrow id := E \left\{ \begin{array}{l} p = \text{lookup}(id.name); \\ \text{if}(p = \text{NULL}) \text{error}(); \\ \text{gen}(:=, E.place, -) \end{array} \right\}$

(2) $E \rightarrow E^{(1)} + E^{(2)} \left\{ \begin{array}{l} E.place() = \text{newtemp}(); \\ \text{gen}(+, E^{(1)}.place, E^{(2)}.place) \end{array} \right\}$

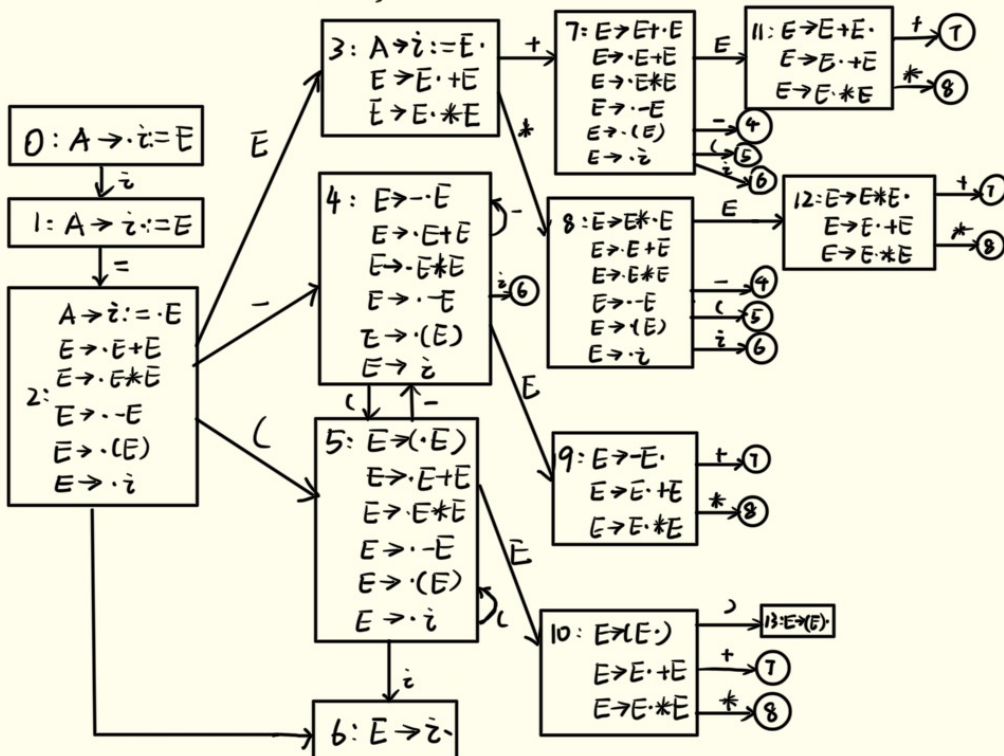
(3) $E \rightarrow E^{(1)} * E^{(2)} \left\{ \begin{array}{l} E.place() = \text{newtemp}(); \\ \text{gen}(*, E^{(1)}.place, E^{(2)}.place) \end{array} \right\}$

(4) $E \rightarrow -E^{(1)} \left\{ \begin{array}{l} E.place() = \text{newtemp}(); \\ \text{gen}(\text{uminus}, E^{(1)}.place, -) \end{array} \right\}$

(5) $E \rightarrow (E^{(1)}) \left\{ E.place = E^{(1)}.place \right\}$

(6) $E \rightarrow id \left\{ \begin{array}{l} p = \text{lookup}(id.name); \\ \text{if}(p \neq \text{NULL}) E.place = \text{Entry}(id); \\ \text{else error}(); \end{array} \right\}$

DFA:



SLR(1)分析表:

Action

Goto

S	:	=	+	*	-	(	)	ε	#	A	E
0								S ₁			
1	S ₂										
2					S ₄	S ₅		S ₆			3
3			S ₇	S ₈					accept		
4					S ₄	S ₅		S ₆			9
5					S ₄	S ₅		S ₄			10
6			r ₆	r ₆			r ₆		r ₆		
7					S ₄	S ₅		S ₄			11
8					S ₄	S ₅		S ₄			12
9			r ₄	r ₄			r ₄		r ₄		
10			S ₇	S ₃			S ₁₂				
11			r ₂	S ₃			r ₂		r ₂		
12			r ₂	r ₂			r ₂		r ₂		
13			r ₅	r ₅			r ₅		r ₅		

产生三元式

符号栈	输入串	动作	三元式
#	B*(-C+D)	移进B	
#B	*(-C+D)	规约为E→id	
#E	*(-C+D)	移进*	
#E*	(-C+D)	移进(	
#E*(	-C+D)	移进-	
#E*(-	C+D)	移进C	
#E*(-C	+D)	规约为E→id	
#E*(-E	+D)	规约为E→-E	T ₁ = -C
#E*(E	+D)	移进+	
#E*(E+	D)	移进D	
#E*(E+D	)	规约为E→id	
#E*(E+E	)	规约为E→E+E	T ₂ = T ₁ + D
#E*(E	)	移进)	
#E*(E)	#	规约为E→(E)	
#E*(E	#	规约为E→*E	T ₃ = B * T ₂
#E*(E	#	赋值规约	A = T ₃

题3:

- 3、写出布尔式 $A \text{ or } (B \text{ and not } (C \text{ or } D))$ 的四元式序列的生成过程。

SDT: (1) $E \rightarrow E^{(1)} \text{ or } E^{(2)}$ $\left\{ \begin{array}{l} E.\text{place} = \text{newtemp}(); \\ \text{Gen}(\text{or}, E^{(1)}.\text{place}, E^{(2)}.\text{place}, E.\text{place}); \end{array} \right\}$

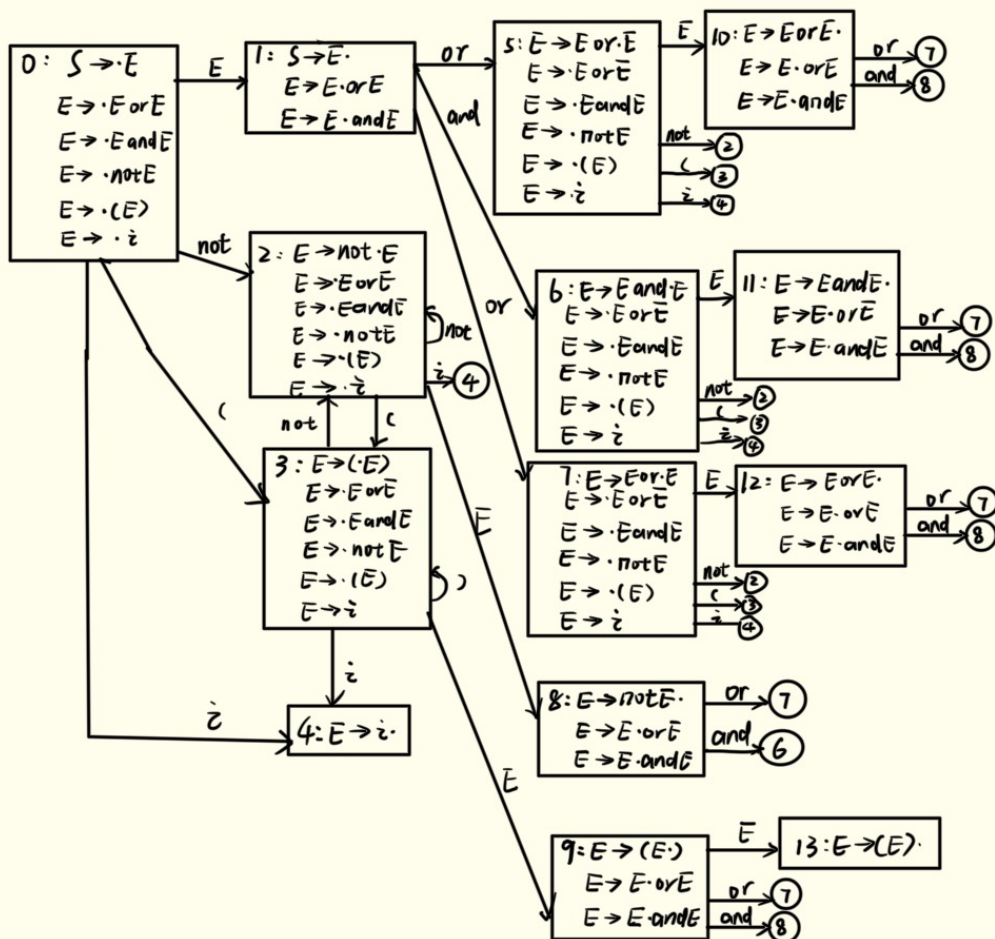
(2) $E \rightarrow E^{(1)} \text{ and } E^{(2)}$ $\left\{ \begin{array}{l} E.\text{place} = \text{newtemp}(); \\ \text{Gen}(\text{and}, E^{(1)}.\text{place}, E^{(2)}.\text{place}, E.\text{place}); \end{array} \right\}$

(3) $E \rightarrow \text{not } E^{(1)}$ $\left\{ \begin{array}{l} E.\text{place} = \text{newtemp}(); \\ \text{Gen}(\text{not}, E^{(1)}.\text{place}, -, E.\text{place}); \end{array} \right\}$

(4) $E \rightarrow (E^{(1)})$ $\{ E.\text{place} = E^{(1)}.\text{place}; \}$

(5) $E \rightarrow \text{id}$ $\left\{ \begin{array}{l} p = \text{lookup}(\text{id}.\text{name}); \\ \text{if } (p \neq \text{NULL}) E.\text{place} = \text{Entry}(\text{id}); \\ \text{else error}(); \end{array} \right\}$

DFA:



SLR(1)分析表

ACTION

GOTO

S	or	and	not	(	)	i	#	E
0			S ₂	S ₃		S ₄		1
1	S ₅	S ₆					accept	
2			S ₂	S ₃		S ₄		8
3			S ₂	S ₃		S ₄		9
4	r ₅	r ₅			r ₅		r ₅	
5			S ₂	S ₃		S ₄		10
6			S ₂	S ₃		S ₄		11
7			S ₂	S ₃		S ₄		12
8	r ₃	r ₃			r ₃		r ₃	
9	S ₇	S ₆			S ₁₃			
10	r ₁	S ₆			r ₁		r ₁	
11	r ₂	r ₂			r ₂		r ₂	
12	r ₁	S ₆			r ₁		r ₁	
13	r ₄	r ₄			r ₄		r ₄	

生成过程:

状态栈	符号栈	输入串	动作	Place	四元式
0	#	A or (B and not (C or D)) #	S ₄		
0 4	# (	or (B and not (C or D)) #	r ₅	A	
0 1	# E	or (B and not (C or D)) #	S ₅	A	
0 1 5	# E or	(B and not (C or D)) #	S ₃	A -	
0 1 5 3	# E or (	B and not (C or D)) #	S ₄	A --	
0 1 5 3 4	# E or ((	and not (C or D)) #	r ₅	A -- B	
0 1 5 3 9	# E or (E	and not (C or D)) #	S ₆	A -- B	
0 1 5 3 9 6	# E or (E and	not (C or D)) #	S ₂	A -- B -	
0 1 5 3 9 6 2	# E or (E and not	(C or D)) #	S ₃	A -- B --	
0 1 5 3 9 6 2 3	# E or (E and not (	C or D)) #	S ₄	A -- B ---	
0 1 5 3 9 6 2 3 4	# E or (E and not ((	or D)) #	r ₅	A -- B --- C	
0 1 5 3 9 6 2 3 9	# E or (E and not (E	or D)) #	S ₇	A -- B --- C	
0 1 5 3 9 6 2 3 9 7	# E or (E and not (E or	D)) #	S ₄	A -- B --- C -	
0 1 5 3 9 6 2 3 9 7 4	# E or (E and not (E or (	) #	r ₅	A -- B --- C - D	
0 1 5 3 9 6 2 3 9 7 1 2	# E or (E and not (E or E	) #	r ₁	A -- B --- C - D	(or, C, D, T ₁)

015396239	#E or (E and not (E	)#	S ₁₃	A--B---T ₁	
015396239 13	#E or (E and not (E)	>#	r ₄	A--B---T ₁	
01539628	#E or (E and not E	>#	r ₃	A--B---T ₁	(not, T ₁ , ..., T ₂)
015396 11	#E or (E and E	>#	r ₂	A--B---T ₂	(and, B, T ₂ , T ₃)
01539	#E or (E	>#	S ₁₃	A--T ₃	
01539 13	#E or (E)	#	r ₄	A--T ₃	
015 10	#E or E	#	r ₁	A--T ₃	(or, A, T ₃ , T ₄)
01	#E	#	accept	T ₄	